

Date	03 July 2026
Team ID	SWTID-2026-4593
Team Size	3
Team Leader	Hari Bellamkonda
Team Member 1	Kesha Vardhan Chinni
Team Member 2	Akili Venkata Siva Sai Srikar
Project Name	Machine Learning-Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Project Planning

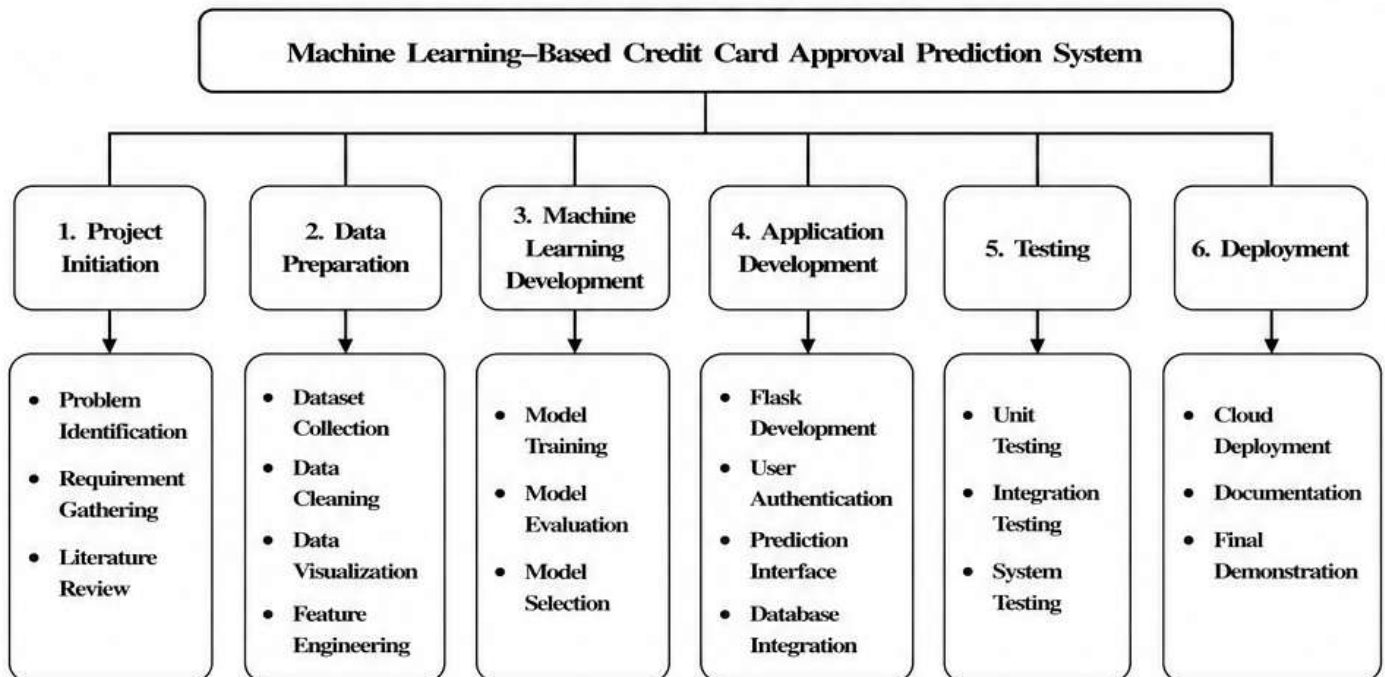
Introduction

The Project Planning document defines the overall execution strategy for developing the Machine Learning-Based Credit Card Approval Prediction System. It identifies project phases, major activities, deliverables, team responsibilities, and the planned development sequence required for successful project completion.

Project Objectives

- Develop a secure Machine Learning-based credit card approval prediction system.
- Automate applicant eligibility evaluation.
- Achieve high prediction accuracy using supervised learning algorithms.
- Build a responsive Flask web application.
- Store applicant and prediction data securely.
- Deploy a scalable solution suitable for real-world banking applications.

Work Breakdown Structure (WBS)



Conclusion

The Work Breakdown Structure provides a systematic approach for executing the project by dividing the complete development process into manageable phases, improving planning, coordination, and project monitoring.

Project Schedule and Resource Planning

The following schedule outlines the planned development activities, estimated duration, project milestones, resource allocation, and potential project risks associated with the Machine Learning–Based Credit Card Approval Prediction System. The plan ensures systematic execution, efficient resource utilization, timely completion, and successful deployment of the proposed solution.

1. Project Timeline

Phase	Activity	Estimated Duration	Deliverable
1	Requirement Analysis	1 Week	Software Requirement Specification (SRS)
2	Dataset Collection and Analysis	1 Week	Prepared and Verified Dataset
3	Data Preprocessing and Feature Engineering	1 Week	Cleaned and Processed Dataset
4	Machine Learning Model Development	2 Weeks	Trained and Evaluated Prediction Models
5	Flask Web Application Development	2 Weeks	Functional Web Application
6	System Testing and Validation	1 Week	Testing and Validation Report
7	Deployment and Documentation	1 Week	Deployed Application and Final Project Documentation

2. Major Milestones

Milestone	Expected Outcome
Requirement Analysis Completed	Approved Project Requirements
Dataset Prepared	Clean Dataset Ready for Training
Machine Learning Model Completed	Best Prediction Model Selected
Web Application Developed	Prediction System Successfully Integrated
System Testing Completed	Verified Functional System
Deployment Completed	Application Successfully Hosted and Ready for Demonstration

3. Risk Assessment

Risk	Mitigation Strategy
Poor Data Quality	Perform thorough preprocessing, validation, and data cleaning.
Low Prediction Accuracy	Train multiple Machine Learning algorithms and select the best-performing model.
Deployment Failure	Test deployment thoroughly using Docker containers before production deployment on Render.
Security Vulnerabilities	Implement secure authentication, encrypted communication, and protected database access.
Project Schedule Delays	Monitor milestones regularly and distribute development tasks efficiently among team members.

4. Resource Allocation

Resource	Technology	Purpose
Python	Python 3.x	Machine Learning model development and backend programming.
Backend Framework	Flask	Develops REST endpoints, business logic, authentication, and prediction services.
Frontend	HTML5, CSS3, JavaScript, Bootstrap 5	Builds a responsive and user-friendly web interface.
Database	MySQL	Stores applicant information, user accounts, application records, and prediction history securely.
Machine Learning Libraries	Scikit-learn, Pandas, NumPy	Supports data preprocessing, feature engineering, model training, evaluation, and prediction.
Development Tools	Visual Studio Code, Jupyter Notebook, Jolbib, Postman	Provides application development, Machine Learning experimentation, model serialization, and API testing.
Version Control	Git and GitHub	Manages source code, project collaboration, version tracking, and repository maintenance.
Cloud Hosting	Render	Hosts the Flask web application with reliable cloud infrastructure and scalable deployment.
Containerization	Docker	Packages the application into portable containers for consistent development, testing, and deployment across environments.

Conclusion

The project planning strategy provides a structured roadmap for successfully implementing the Machine Learning–Based Credit Card Approval Prediction System. By defining project phases, timelines, milestones, resource allocation, deployment strategy, and risk management practices, the project ensures efficient development, accurate prediction performance, secure data handling, and reliable cloud deployment.